

Decimal adders \ Decoders

$$12 = 8 + 4 + 2(0) + 1(0)$$

$$12 = 1100$$



$$\underline{0001} \ 0010 = \text{BCD}$$

① Identify inputs/outputs for a BCD Adder

- single BCD = 4 bits

- 2 BCD = 8 bits

- We also need a carry-in from the previous position

BCD adder needs 9 inputs

Outputs for BCD adder

- 4 bits for output sum

- we also need a carry out, which is another 1 bit

BCD adder results in 5 outputs.

To build a BCD Adder:

- we can reuse a 4-bit binary adder
- fix up the answer if it is illegal

2 decimal digits: $\left. \begin{array}{r} 9 \\ + 9 \\ + 1 \\ \hline \end{array} \right\} 19 = \begin{array}{r} 1001 \\ + 1001 \\ + 0001 \\ \hline 10011 \end{array} \left. \vphantom{\begin{array}{r} 9 \\ + 9 \\ + 1 \\ \hline \end{array}} \right\} \text{Raw Sum}$

Bit₄ Bit₃ Bit₂ Bit₁ Bit₀

$$16 + 8 + 4 + 2 + 1 = 31$$

BCD Adder: sum can be equivalent to any values in the range 0-31.

carry out

K	Binary Sum					BCD Sum				Decimal
	Z₈	Z ₄	Z ₂	Z ₁	C	S ₈	S ₄	S ₂	S ₁	
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	2
0	0	0	1	1	0	0	0	1	1	3
0	0	1	0	0	0	0	1	0	0	4
0	0	1	0	1	0	0	1	0	1	5
0	0	1	1	0	0	0	1	1	0	6
0	0	1	1	1	0	0	1	1	1	7
0	1	0	0	0	0	1	0	0	0	8
0	1	0	0	1	0	1	0	0	1	9
0	1	0	1	0	1	0	0	0	0	10
0	1	0	1	1	1	0	0	0	1	11
0	1	1	0	0	1	0	0	1	0	12
0	1	1	0	1	1	0	0	1	1	13
0	1	1	1	0	1	0	1	0	0	14
0	1	1	1	1	1	0	1	0	1	15
1	0	0	0	0	1	0	1	1	0	16
1	0	0	0	1	1	0	1	1	1	17
1	0	0	1	0	1	1	0	0	0	18
1	0	0	1	1	1	1	0	0	1	19

Whenever the sum is in illegal form, add 6 (0110) to the raw sum

Raw Sum

Ex) 7+8 in BCD

$$7 = 0111$$

$$8 = 1000$$

$$+ 0000 \leftarrow \text{carry-in}$$

$$1111 = 15 \text{ in binary}$$

$$01111 \left\{ \begin{array}{l} K=0 \\ 2=1111 \end{array} \right.$$

$$\text{To fix: } \begin{array}{r} 110 \\ 1111 \end{array}$$

$$+ 0110$$

$$10101 \rightarrow \boxed{0001 \ 0101}$$

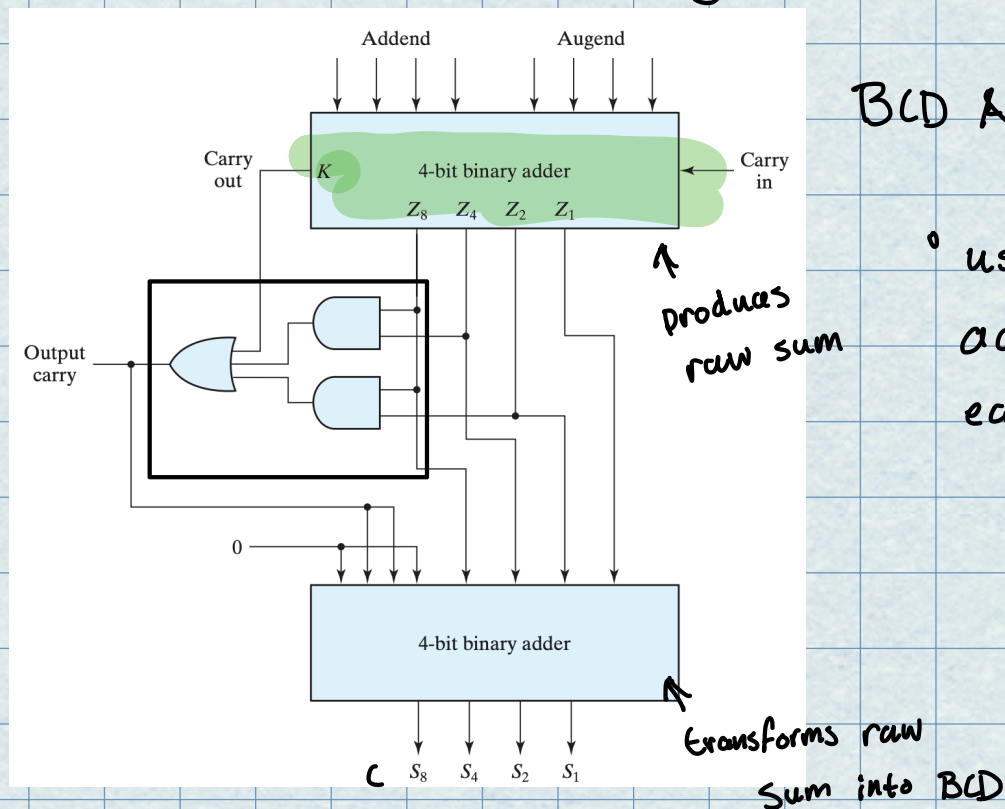
Conditions for when to add 0110 in the adder:

① If $K=1$ (16, 17, 18, 19 means always correct)

② If $K=0$, on the condition that the value is 10-15, Z_8 must also be =1.

③ Among the codes where $Z_8=1$, the illegal ones (10-15) all have either $Z_4=1$ or $Z_2=1$.

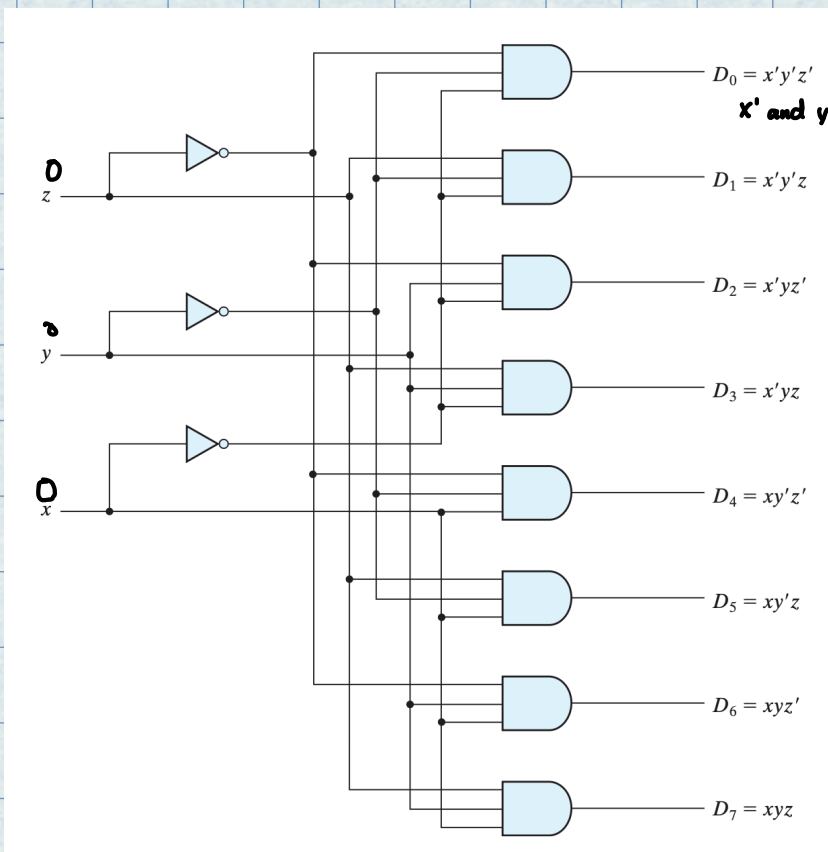
Condition: $C = K + Z_8 Z_4 + Z_8 Z_2$
 $C=1 \rightarrow$ add 0110
 $C=0 \rightarrow$ do nothing



BCD Adder:

• uses TWO 4-bit binary adders stacked on top of each other

Decoder: a combinational circuit with n input lines and 2^n output lines. For any pattern, exactly one output line goes active. The others stay inactive.

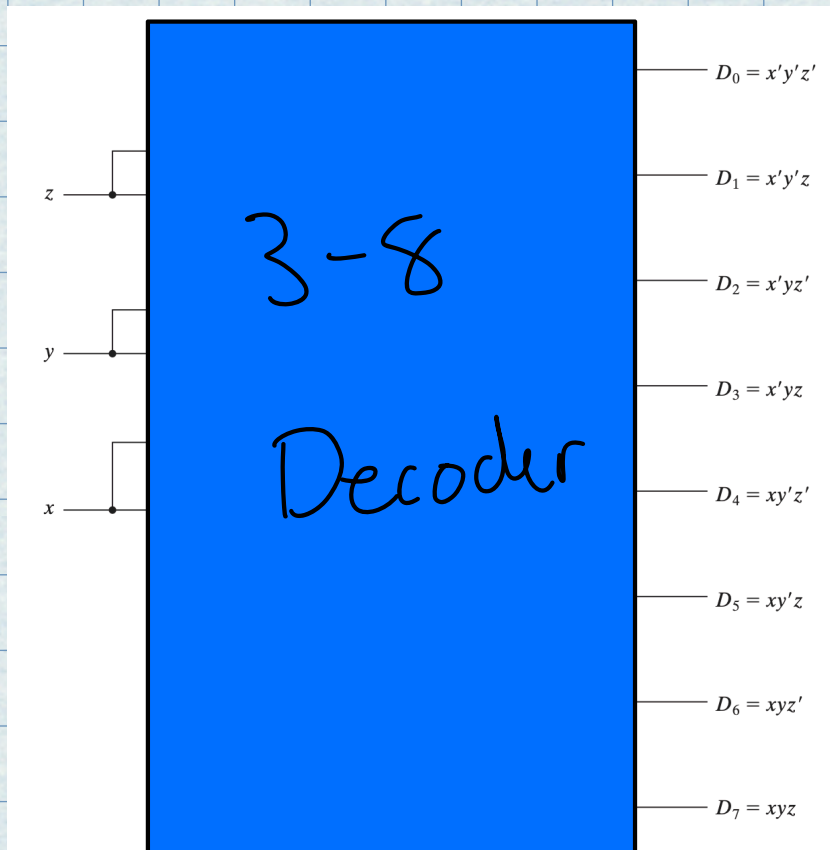


3 inputs (x, y, z)

8 outputs ($2^3 = 8$)

3-8 line Decoder

Inputs			Outputs							
x	y	z	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1



Because a decoder generates every minterm, and because boolean functions can be written in sum of minterms, any combinational circuit can be implemented using a decoder and or gates.

Recipe:

- ① Write each output as a sum of minterms
- ② Pick a decoder (n -to- 2^n decoder)
- ③ For each output, OR together the decoder lines listed in the outputs minterm list

Ex) Adder: Sum $\rightarrow S \rightarrow S(X, Y, Z) = \Sigma(1, 2, 4, 7)$
Carry $\rightarrow C \rightarrow C(X, Y, Z) = \Sigma(3, 5, 6, 7)$

3-8 decoder: Output S = OR of decoder output
Output C = OR of decoder output lines.

